SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS **COURSE CODE: MLA01**

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.*(BTL 3)

Assessment Tool Weightage: 15%

Total Marks: 100 **Time: 60 Mins**

No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I K.Hemakshitha(192425088) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	

8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

ANSWER ANY ONE OF THE FOLLOWING

Q.No	Use Case Scenario	
1	**Smart Healthcare Diagnosis**	A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention. Tasks: (i) Identify facts, entities, predicates, variables, and constants. (ii) Represent the knowledge using Propositional Logic. (iii) Represent the knowledge using FOL with suitable quantifiers. (iv) Perform unification for two relevant predicates. (v) Apply resolution to prove whether Ravi has a viral infection. (vi) Show the complete inference sequence and final conclusion.

ANSWER

1. Smart Healthcare Diagnosis

Problem Statement

A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention.

Tasks

(i) Identify facts, entities, predicates, variables, and constants.

(ii) Represent the knowledge using Propositional Logic.

(iii) Represent the knowledge using FOL with suitable quantifiers.

(iv) Perform unification for two relevant predicates.

(v) Apply resolution to prove whether Ravi has a viral infection.

(vi) Show the complete inference sequence and final conclusion.

(i) Facts, Entities, Predicates, Variables and Constants

Entities

- Ravi
- Patients
- Flu
- Viral infection
- Medical attention

Facts

- Ravi has fever.
- Ravi has cough.
- Ravi has body pain.

Predicates

- Fever(x) : x has fever.
- Cough(x) : x has cough.
- BodyPain(x) : x has body pain.
- Flu(x) : x has flu.
- ViralInfection(x) : x has viral infection.
- MedicalAttention(x) : x requires medical attention.

Variables

- x represents any patient.

Constants

- Ravi represents the specific patient Ravi.

Rules

1. A patient with fever and cough may have flu.
2. A patient with flu and body pain may have viral infection.
3. A patient with viral infection requires medical attention.

(ii) Propositional Logic Representation

Let:

- F = Ravi has fever.
- C = Ravi has cough.
- B = Ravi has body pain.
- L = Ravi has flu.
- V = Ravi has viral infection.
- M = Ravi requires medical attention.

Facts

F

C

B

Rules

Rule 1:

(F AND C) implies L

Rule 2:

(L AND B) implies V

Rule 3:

V implies M

Complete Propositional Knowledge Base
F AND C AND B
(F AND C) implies L
(L AND B) implies V
V implies M
Therefore, the knowledge base allows us to derive:
F AND C gives L
L AND B gives V
V gives M

(iii) First-Order Logic Representation
Facts
Fever(Ravi)
Cough(Ravi)
BodyPain(Ravi)
Rule 1
All patients who have fever and cough may have flu.
For every x, (Fever(x) AND Cough(x)) implies Flu(x)
Rule 2
All patients who have flu and body pain may have viral infection.
For every x, (Flu(x) AND BodyPain(x)) implies ViralInfection(x)
Rule 3
All patients with viral infection require medical attention.
For every x, ViralInfection(x) implies MedicalAttention(x)
Quantifiers Used
The universal quantifier For every x is used because the rules apply to any patient.
FOL Knowledge Base

1. Fever(Ravi)
2. Cough(Ravi)
3. BodyPain(Ravi)
4. For every x, (Fever(x) AND Cough(x)) implies Flu(x)
5. For every x, (Flu(x) AND BodyPain(x)) implies ViralInfection(x)
6. For every x, ViralInfection(x) implies MedicalAttention(x)

(iv) Unification and Substitution
Consider the following two relevant predicates:
Fever(x)
Fever(Ravi)
For these two predicates to match:
x = Ravi
Therefore, the substitution is:
{x / Ravi}
After applying the substitution:
Fever(x) becomes Fever(Ravi)

Hence, the two predicates are successfully unified.
Another Relevant Unification
Consider:
ViralInfection(x)
ViralInfection(Ravi)
The required substitution is:
{x / Ravi}
Therefore:
ViralInfection(x) becomes ViralInfection(Ravi).
Most General Unifier
MGU = {x / Ravi}
Thus, the variable x is replaced by the constant Ravi.

(v) Resolution to P
rove Ravi Has Viral Infection
Step 1: Convert the Rules into Clause Form
Rule 1:
(Fever(x) AND Cough(x)) implies Flu(x)
Clause form:
NOT Fever(x) OR NOT Cough(x) OR Flu(x)
Rule 2:
(Flu(x) AND BodyPain(x)) implies ViralInfection(x)
Clause form:
NOT Flu(x) OR NOT BodyPain(x) OR ViralInfection(x)
Rule 3:
ViralInfection(x) implies MedicalAttention(x)
Clause form:
NOT ViralInfection(x) OR MedicalAttention(x)
Facts
Fever(Ravi)
Cough(Ravi)
BodyPain(Ravi)
Step 2: Negate the Goal
The goal is:
ViralInfection(Ravi)
For resolution by contradiction, assume the opposite:
NOT ViralInfection(Ravi)

Step 3: Derive Flu(Ravi)
Use:
NOT Fever(x) OR NOT Cough(x) OR Flu(x)
and the fact:
Fever(Ravi)
Using substitution:

{x / Ravi}
we obtain:
NOT Cough(Ravi) OR Flu(Ravi)
Now use:
Cough(Ravi)
Resolution gives:
Flu(Ravi)
Therefore:
Flu(Ravi) is proved.

Step 4: Derive ViralInfection(Ravi)
Use:
NOT Flu(x) OR NOT BodyPain(x) OR ViralInfection(x)
with:
Flu(Ravi)
and:
BodyPain(Ravi)
Using:
{x / Ravi}
First resolution gives:
NOT BodyPain(Ravi) OR ViralInfection(Ravi)
Using BodyPain(Ravi), resolution gives:
ViralInfection(Ravi)
Therefore:
ViralInfection(Ravi) is proved.

Step 5: Resolve with the Negated Goal
We have:
ViralInfection(Ravi)
and the negated goal:
NOT ViralInfection(Ravi)
Resolution produces the empty clause:
□
The empty clause represents a contradiction.
Therefore, the negated goal is false, and the original goal is true.

(vi) Complete Inference Sequence and Final Conclusion
Inference Sequence
Step 1
Given:
Fever(Ravi)
Cough(Ravi)
Using the rule:
Fever(x) AND Cough(x) implies Flu(x)

Conclusion:
Flu(Ravi)
Step 2
Given:
Flu(Ravi)
BodyPain(Ravi)
Using the rule:
Flu(x) AND BodyPain(x) implies ViralInfection(x)
Conclusion:
ViralInfection(Ravi)
Step 3
Given:
ViralInfection(Ravi)
Using the rule:
ViralInfection(x) implies MedicalAttention(x)
Conclusion:
MedicalAttention(Ravi)

Mapping / Logic Diagram

```
Fever(Ravi) + Cough(Ravi)
           |
           |
           v
       Flu(Ravi)
           |
           |
   + BodyPain(Ravi)
           |
           v
  ViralInfection(Ravi)
           |
           |
           v
 MedicalAttention(Ravi)
```

Final Conclusion
Based on the given facts and rules, Ravi has fever, cough, and body pain. Fever and cough allow us to infer that Ravi has flu. Flu combined with body pain allows us to infer that Ravi has a viral infection. Since viral infection requires medical attention, Ravi also requires medical attention.
Therefore, the final conclusion is:
ViralInfection(Ravi) is TRUE.
MedicalAttention(Ravi) is TRUE.
The resolution proof also produces the empty clause after combining the derived ViralInfection(Ravi) with its negation. Hence, the conclusion that Ravi has a viral infection is logically proved.

EVALUATION RUBRICS:

Criterion	**Weightage (%)**	**Excellent**	**Good**	**Average**	**Poor**
Problem Understanding & Knowledge Identification	15%	13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10–12% – Identifies most relevant knowledge with minor omissions.	7–9% – Identifies basic knowledge but misses several important elements.	0–6% – Poor or incorrect identification of the given knowledge.
Logical Representation & Mapping	20%	17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.	13–16% – Provides mostly correct logical mappings with minor errors.	9–12% – Provides partial mappings with several logical errors.	0–8% – Logical mapping is largely incorrect or missing.
Predicates, & Variables, Constants Quantifiers	15%	13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% – Uses the concepts correctly with minor errors.	7–9% – Demonstrates partial understanding with some incorrect usage.	0–6% – Concepts are incorrectly applied or missing.

Logical Connectives & Formula Construction	10%	9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to construct valid logical formulas.	7–8% – Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.
Inference & Logical Reasoning	15%	13–15% – Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.	10–12% – Reasoning is mostly correct with minor gaps.	7–9% – Provides partial reasoning with some incorrect conclusions.	0–6% – Reasoning is incorrect, unsupported, or missing.
Unification & Substitution	10%	9–10% – Correctly identifies substitutions and Most General Unifier (MGU) and applies them accurately.	7–8% – Correctly performs most unification steps with minor errors.	5–6% – Demonstrates basic understanding but has several errors.	0–4% – Unable to perform unification correctly.
Resolution / Proof Construction	10%	9–10% – Correctly applies resolution and presents a clear, logically valid proof of the query.	7–8% – Resolution is mostly correct with minor errors.	5–6% – Shows partial understanding but proof contains gaps.	0–4% – Resolution is incorrect or not demonstrated.

Clarity, & Mapping Diagram Presentation	5%	5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation.	4% – Presentation is clear with minor formatting or diagram issues.	3% – Basic presentation with limited clarity.	0–2% – Poorly organized, unclear, or incomplete presentation.